

Q. How is the performance of a computer system measured? Discuss CPU Execution Time, Clock Rate, and the various factors that affect computer performance.

📌 Definition to Remember

Computer **Performance** measures how quickly a system executes tasks. It is inversely proportional to execution time ($\text{Performance} = 1 / (\text{Execution Time})$). The primary metric is **CPU Execution Time**, which depends on Instruction Count, CPI (Cycles Per Instruction), and Clock Rate.

1. CPU Execution Time

The actual time the CPU spends computing a specific task (excludes I/O wait time).

Formulas:

* $\text{CPU Execution Time} = \text{CPU Clock Cycles} \times \text{Clock Cycle Time}$

* $\text{CPU Execution Time} = \text{CPU Clock Cycles} / (\text{Clock Rate})$

2. Clock Rate and Clock Cycle

- **Clock Cycle (Period):** The fundamental unit of time in a computer (e.g., 1 ns), generated by a hardware oscillator.
- **Clock Rate (Frequency):** The inverse of cycle time. Number of cycles per second, measured in **Hertz (Hz)** (e.g., 3 GHz). Higher clock rate generally means faster execution.

3. Other Key Performance Metrics

- **CPI (Cycles Per Instruction):** Average number of clock cycles needed to execute one instruction.
- $\text{Execution Time} = \text{Instruction Count} \times \text{CPI} \times \text{Clock Cycle Time}$
- **MIPS (Millions of Instructions Per Second):** A metric expressing execution speed.
- $\text{MIPS} = \text{Instruction Count} / (\text{Execution Time} \times 10^6)$

4. Factors Affecting Computer Performance

Factor	Impact on Performance
Clock Speed	Higher clock speed executes instructions in less time.
Compiler Efficiency	Better compilers generate fewer machine instructions (lowers Instruction Count).
Architecture / CPI	Pipelining, superscalar, and multi-core designs lower the average CPI.
Cache Memory	Fast cache reduces time the CPU spends waiting for data from main memory.
Bus Width	A wider data bus transfers more data per cycle, reducing memory bottleneck.

★ Must-Write Points (for 10 marks)

1. Performance is inversely proportional to execution time: $\text{Perf} = 1 / \text{Time}$.
2. **CPU Execution Time** = (Instruction Count) $\times$ (CPI) $\times$ (Clock Cycle Time).

3. **Clock Cycle**: Fundamental time unit; **Clock Rate**: Cycles per second (Hz).
4. **CPI**: Cycles Per Instruction. Lower CPI means better architecture (via pipelining).
5. **MIPS**: Millions of Instructions Per Second (measures execution speed rate).
6. Compiler efficiency affects the total Instruction Count.
7. Cache memory and Bus Width heavily impact performance by reducing data fetch delays.

⚡ Quick Recall

Performance = $1/\text{Execution Time}$ → Exec Time = $(\text{Instruction Count} \times \text{CPI}) / \text{Clock Rate}$ → CPI = Cycles per Instruction → Factors: Clock Speed, Compiler, Pipelining (CPI), Cache, Bus Width